

Q3 2023 Earnings Call

Company Participants

- Farhan Ahmad, Vice President, Investor Relations
- Mark Murphy, Executive Vice President and Chief Financial Officer
- Sanjay Mehrotra, President and Chief Executive Officer

Other Participants

- Ambrish Srivastava, BMO Capital Markets
- C.J. Muse, Evercore ISI
- Harlan Sur, J.P. Morgan
- Krish Sankar, Cowen
- Timothy Arcuri, UBS
- Tom O'Malley, Barclays

Presentation

Operator

Thank you for standing by and welcome to Micron's Third Quarter 2023 Financial Call. At this time, all participants are in listen-only mode. After the speakers' presentation, there will be a question-and-answer session. (Operator Instructions) As a reminder, today's program is being recorded.

And now I'd like to introduce your host for today's program, Farhan Ahmad, Vice President, Investor Relations. Please go ahead, sir.

Farhan Ahmad {BIO 20748275 <GO>}

Thank you, and welcome to Micron Technology's fiscal third quarter 2023 financial conference call. On the call with me today are Sanjay Mehrotra, our President and CEO; and Mark Murphy, our CFO. Today's call is being webcast from our Investors Relations site at investors.micron.com, including audio and slides.

In addition, the press release detailing our quarterly results has been posted on our website, along with the prepared remarks for this call. Today's discussion of the financial results is presented on a non-GAAP financial basis, unless otherwise specified. A reconciliation of GAAP to non-GAAP financial measures can be found on our website. We encourage you to visit our website at micron.com throughout the quarter for the most current information on the company, including information on the financial conferences that we may be attending. You can also follow us on Twitter, @MicronTech.

As a reminder, the matters we are discussing today include forward-looking statements regarding market demand and supply, our expected results and other matters. These forward-looking statements are subject to risks and uncertainties that may cause actual results to differ materially from the statements made today.

We refer you to the documents we filed with the SEC, including our most recent Form 10-K and 10-Q for a discussion of the risks that may affect our future results. Although we believe that the expectations reflected in the forward-looking statements are reasonable, we cannot guarantee

future results, levels of activity, performance or achievements. We are under no duty to update any of the forward-looking statements to conform these statements to actual results.

I'll now turn the call over to Sanjay.

Sanjay Mehrotra {BIO 1977912 <GO>}

Thank you, Farhan. Good afternoon, everyone. Micron delivered fiscal third quarter revenue within our guidance range with gross margin and EPS above the range. The ongoing improvement of customer inventories and memory content growth are driving higher industry demand while production cuts across the industry continue to help reduce excess supply.

As a result, pricing trends are improving and we have increased confidence that the industry has passed the bottom for both quarterly revenue and year-on-year revenue growth. Our technology leadership and strengthening product portfolio position us well across diverse growth markets, including AI, and memory-centric computing. Beyond this downturn, we expect to see record TAM in calendar 2025 along with a return to more normalized levels of profitability.

The impact of the May 21st decision by the Cyberspace Administration of China on Micron's business remains uncertain and fluid. Several Micron customers, including mobile OEMs, have been contacted by certain Critical Information Infrastructure operators or representatives of the government in China, concerning the future use of Micron products.

As discussed before, Micron's revenue with companies headquartered in mainland China and Hong Kong, including direct sales as well as indirect sales through distributors, accounts for approximately a quarter of Micron's worldwide revenue, and remains the principal exposure. We currently estimate that approximately half of that China headquartered customer revenue, which equates to a low double-digit percentage of Micron's worldwide revenue is at risk of being impacted. This significant headwind is impacting our outlook and slowing our recovery. Micron is working to mitigate this impact over time and expects increased quarter-to-quarter revenue variability. Micron's long-term goal is to retain its worldwide DRAM and NAND share.

Turning to technology. Micron continues to lead the industry in both DRAM and NAND technology. We are investing prudently to maintain our technology competitiveness while managing CapEx, node ramps, and wafer start reductions to reduce our bit supply and align it with demand.

Our industry-leading 1-beta DRAM and 232-layer NAND nodes are achieving world-class yields, and the yield ramp for these nodes has been faster than any of our prior nodes. These leadership nodes provide a strong cost capability along with best-in-class power and performance specifications that will be leveraged across the portfolio of DRAM and NAND products.

In Fiscal Q3, we achieved several important product qualifications on these advanced nodes and are well positioned to ramp them in fiscal 2024. We are also making good progress towards the introduction of our EUV-based 1-gamma node in 2025. This node will be manufactured first in our Taiwan site where we already have EUV capability installed and operating in preparation for this ramp. We also recently announced plans to bring EUV technology to our fab in Hiroshima, Japan with support from the Japanese government. Micron will be the first company to bring EUV technology to Japan for production.

We are also advancing our global assembly and test network in order to support our product portfolio and extend our ability to deliver on global customer demand in the future. In China, we announced an investment of approximately \$600 million over the next several years in our

operations in Xi'an. This builds on our long history of significant investment in our Xi'an assembly packaging and test operations.

As a part of this investment, we have decided to purchase the assembly equipment of our partner, Powertech Semiconductor Xi'an, who has been operating inside our Xi'an facility for the last eight years. We also intend to construct a new building at our Xi'an site to provide space to add more product capabilities. This will allow us, over time, to serve more of the demand from our customers in China from the Xi'an site.

In India, with the strong support of the Indian government, we will build a new assembly and test facility in Gujarat to address demand for the latter half of this decade. We are also increasing our investments in assembly and test capacity in Taiwan for high-bandwidth memory products as we gear up for stronger demand in this segment driven by the AI wave.

Now turning to our end markets. Customers continue to make progress in reducing their excess inventory in fiscal Q3. Most customer inventories in the PC and smartphone segments are close to normal levels now, consistent with our forecast six months ago. Some of these inventory levels can get distorted by customer attempts to leverage current prices, which are deemed to be transient and unsustainable at these levels, to purchase additional volumes before prices rise significantly. Data center customer inventory is also improving and will likely normalize around the end of this calendar year or somewhat thereafter, depending on the growth in traditional data center spending.

In data center, we saw strong sequential revenue growth in both cloud and enterprise in fiscal Q3, driven by some recovery from depressed sales levels in fiscal Q2. The recent acceleration in the adoption of generative AI is driving higher-than-expected industry demand for memory and storage for AI servers, while traditional server demand for mainstream data center applications continues to be lackluster.

Micron's product portfolio and roadmap of innovative products position us to capture growth opportunities from AI and data-centric computing architectures for both training and inferencing. Increasingly large AI models with an exponentially growing number of parameters are driving demand for dramatically higher memory content.

As we have said before, AI servers have six to eight times the DRAM content of a regular server and three times the NAND content. In fact, some customers are deploying AI compute capability with substantially higher memory content. A striking example is NVIDIA's DGX GH200 supercluster, which shows just how memory intensive AI workloads can be. It provides developers the ability to support giant models with a massive shared memory space of 144 terabyte.

A significant majority of that memory footprint is enabled by a joint development project between our two companies that extends Micron's low-power DRAM leadership to server-class applications. We are proud to pioneer the differentiated LPDRAM innovation to deliver a significant reduction in data center power consumption compared to DDR-based solutions, helping to support our customers' green initiatives.

High-bandwidth memory used in high-performance computing is seeing very strong demand this year, driven by demand for generative AI. We are working closely with our customers and have begun sampling our industry-leading HBM3 product offering. The customer response has been strong, and we believe our HBM3 product delivers significantly higher bandwidth than competing solutions and establishes the new benchmark in performance and power consumption, supported by our 1-beta technology, TSV and other innovations, enabling a differentiated

advanced packaging solution. We expect to begin a mass production ramp for this exciting HBM3 product in early calendar 2024 and to achieve meaningful revenues in fiscal 2024.

Micron also has a strong position in the industry transition to D5, which is the latest generation of DDR memory. Our D5 percentage of DRAM shipments has more than doubled from fiscal Q2 to Q3, and we expect Micron D5 volume to cross over D4 at the end of first calendar quarter of 2024 versus mid-calendar 2024 for the industry.

Micron's 1-alpha D5 modules are qualified and shipping to data center customers. We are also making good progress on our 1-beta-based high-density 128-gigabyte D5 modules using a 32-gigabit die that optimizes cost and performance to provide customers with a lower cost of ownership solution for memory-intensive workloads like AI. We expect these high-density modules to ramp in calendar Q2 of 2024 with significant cost improvements over today's expensive TSV-based solutions in the industry.

Our 96-gigabyte D5 high-density module built on 1-alpha technology using 24-gigabit die is already shipping in volume and delivers equivalent performance for the majority of workloads versus the more expensive TSV dual-die package based 128-gigabyte modules.

In data-centered SSDs, Micron's entire portfolio is now on 176-layer or 232-layer NAND, demonstrating our product and technology leadership. We are in a strong position to serve AI demand for fast storage as these data-intensive applications proliferate.

In fiscal Q3, we launched the world's first 200-plus layer NAND data-centered SSD, and qualification is in progress at multiple key customers to support AI cluster installations. In fact, we have already passed qualification of this product at a critical server OEM partner. We also launched our extreme endurance data centered SSD, which offers superior scalability and affordability versus hard drives.

In PCs, we now forecast calendar 2023 PC unit volume to decline by a low double-digit percentage year-over-year with PC units expected to be below the pre-COVID levels last seen in 2019.

We are excited about the ongoing industry transition to D5 and are well positioned for it with our strong D5 product lineup. Industry client D5 mix is expected to cross over from D4 in early calendar 2024. In fiscal Q3, we achieved record quarterly client SSD bit shipments driven by share growth in client SSDs as customers adopted our industry-leading solutions. Our SSD QLC bit shipment mix reached a new record for the third consecutive quarter with growth in both client and consumer. Last month, we launched Crucial T700, the world's fastest Gen 5 PCIe consumer SSD, built with our 232-layer NAND.

In graphics, industry analysts continue to expect graphics' TAM growth CAGR to outpace the broader market supported by applications across client and data center. We expect customer inventories to normalize in calendar Q3. We plan to introduce our next generation G7 product on our industry-leading 1-beta node in the first half of calendar year 2024.

In mobile, we now expect calendar 2023 smartphone unit volumes to be down by a mid-single-digit percentage year-over-year. While units are weaker, we are seeing stronger memory content growth driven by a mix shift toward premium phones and elasticity. We expect sequential growth in fiscal Q4 as customers prepare for upcoming product launches in the back half of calendar 2023.

In fiscal Q3, we achieved key mobile customer qualifications on our 1-beta based LP5X and started high-volume revenue shipments to Tier 1 OEMs. In addition, we achieved significant milestones in UFS with the qualification and ramp of a high-capacity uMCP5 featuring 16 gigabyte of DRAM and 512 gigabyte of NAND. We have also started to sample a new UFS4 product based on our latest 232-layer NAND technology, which enables industry-leading performance for flagship handsets.

Last, I'll cover the auto and industrial end markets, which represent over 20% of our revenue and contribute more stable revenue and profitability. Micron continues to lead in automotive, which is a key market and growth driver for us. In fiscal Q3, auto revenue reached another quarterly record and grew by a high single-digit percentage year-over-year. We continue to expect growth in auto memory demand for the second half of calendar 2023, driven by easing non-memory semiconductor supply, normalizing customer inventory levels, and increasing memory content per vehicle.

The industrial market saw early signs of recovery in fiscal Q3. Inventory levels are stabilizing at distribution partners and at the majority of our customers. As a result, we expect an improvement in demand in the second half of calendar 2023. We are excited about our growth prospects in this market as industrial customers continue to adopt and implement IoT, AI, and machine learning in the factory.

Now turning to industry outlook. Our expectations for calendar 2023 industry bit demand growth have been further reduced to low to mid-single digits in DRAM and to high single digits in NAND, which are well below the expected long-term CAGR of mid-teens percentage range in DRAM and low 20s percentage range in NAND. While the AI-driven demand has been stronger than our expectations three months ago, the PC, smartphone, and traditional server demand forecasts are now lower.

We continue to expect stronger industry bit shipments for DRAM and NAND in the second half of the calendar year, driven by secular content growth and continued improvement in customer inventory.

While the industry demand forecast for calendar 2023 is now lower, the significant supply reductions across the industry have started to stabilize the market. We see both DRAM and NAND year-over-year supply growth to be negative for the industry in calendar 2023, as utilization and CapEx cuts across the industry impact supply growth. While supply-demand balance is improving, due to the excess inventory, profitability and cash flow will remain extremely challenged for some time. Market recovery can accelerate if there is further reduction in industry production, and these cuts are sustained well into calendar 2024.

In response to the industry environment, Micron has taken decisive actions to bring our supply back in balance with demand. We expect Micron's year-on-year bit supply growth to be meaningfully negative for DRAM. We also expect to produce fewer NAND bits in calendar 2023 than in calendar 2022. Our fiscal 2023 CapEx plan of \$7 billion is down more than 40% from last year, with WFE down more than 50%. We continue to expect fiscal '24 WFE to be down year-on-year. Recently, we have further reduced wafer starts to approach 30% in both DRAM and NAND. We currently expect reduced wafer starts will continue well into calendar 2024 as we remain focused on managing down our inventories and controlling our supply.

I will now turn it over to Mark.

Mark Murphy {BIO 16136048 <GO>}

Thanks, Sanjay. Good afternoon, everyone. Fiscal Q3 results were in line to better than expectations, with revenue coming in above the midpoint of our guidance range and gross margin and EPS exceeding the high end of the range. Total fiscal Q3 revenue was approximately \$3.8 billion, up 2% sequentially and down 57% year-over-year. Fiscal Q3 revenue included \$72 million from an insurance settlement disclosed at the time we provided guidance.

Fiscal Q3 DRAM revenue was \$2.7 billion, representing 71% of total revenue. DRAM revenue declined 2% sequentially, with bit shipments increasing in the 10% range and prices declining by approximately 10%. Fiscal Q3 NAND revenue was \$1 billion, representing 27% of Micron's total revenue. NAND revenue increased 14% sequentially, with bit shipments increasing in the upper 30% range and prices declining in the mid-teens percentage range.

Now turning to revenue by business unit. Compute and Networking Business Unit revenue was \$1.4 billion, up 1% sequentially. Strong sequential growth in server and graphics revenues was offset by a decline in client. Embedded Business Unit revenue was \$912 million, up 5% sequentially.

On a sequential basis, automotive and consumer revenues were strong. Revenue for the Mobile Business Unit was \$819 million, down 13% sequentially due to timing of shipments. As Sanjay mentioned, we expect growth in mobile revenues in fiscal Q4. Revenue for the Storage Business Unit was \$627 million, up 24% sequentially and driven by increased shipments across most of the portfolio. The consolidated gross margin for fiscal Q3 was negative 16%, improving 15 percentage points sequentially. This result was negatively impacted by approximately \$400 million, or 11 percentage points of write-downs associated with inventory produced in the quarter.

Operating expenses in fiscal Q3 were \$866 million, down roughly \$50 million sequentially. OpEx benefited from ongoing expense reduction initiatives and gains on sales of certain assets. We had an operating loss of roughly \$1.5 billion in fiscal Q3, resulting in an operating margin of negative 39%, improved from negative 56% in the prior quarter. Fiscal Q3 taxes were \$102 million, higher than expectations at the time of our guidance, driven by one-time discrete items. As mentioned in previous quarters, despite a consolidated loss on a worldwide basis, we still have taxes payable in certain geographies due to taxable income levels reported in those geographies.

The non-GAAP loss per share in Fiscal Q3 was \$1.43, down from a loss per share of \$1.91 in the prior quarter, and earnings per share of \$2.59 in the prior year. Fiscal Q3 EPS included approximately \$0.37 of losses from the impact of the inventory write-down associated with inventory produced in the quarter.

Turning to cash flows and capital spending. Our operating cash flows were approximately \$24 million. Capital expenditures were \$1.4 billion during the quarter. We continue to expect capital expenditures to be approximately \$7 billion for the fiscal year, thus near \$1 billion in fiscal Q4. Free cash flow is negative \$1.4 billion in the quarter and improved from the previous two quarters.

Our fiscal Q3 ending inventory was \$8.2 billion, or 168 days. Due to increases in process steps and product complexity, we now target inventory levels of around 120 days, which at present would equate to approximately \$6 billion. Our current inventories include strategic stocks of approximately \$1 billion over target levels associated with build-ahead of product for cost optimization and risk mitigation. At quarter end, we held cash and investments of \$11.4 billion and had total liquidity of \$13.9 billion, including our untapped credit facility.

We issued \$1.5 billion of long-term debt in the quarter and with part of those proceeds, paid down \$600 million of our term loan facility, resulting in a net increase to debt of \$900 million. Our

fiscal Q3 ending debt was \$13.2 billion.

Now turning to our outlook for the fiscal fourth quarter. As mentioned in filings and our comments today, the CAC decision is a headwind to our outlook. We expect the revenue impact to vary by quarter, with the impact in fiscal Q4 being less than the quarterly impact in the first half of fiscal 2024. Over time, we have a goal of retaining our global market share in both DRAM and NAND.

In fiscal Q4, as the industry demand continues to improve, and despite the effects on our business from the CAC decision, we still see record bit shipments. Fiscal Q4 gross margin will be impacted by costs from underutilization, weak pricing levels, and product mix. In the current business environment, the gap between our DRAM and NAND profitability is significant, and changes in the mix can drive large variability in gross margins.

Our gross margin guidance does not contemplate additional write downs of inventory. We continue to aggressively manage our operating expenses and remain on track to exit the fiscal year at less than \$850 million. Looking beyond fiscal Q4, we expect OpEx to increase over \$50 million in fiscal Q1 2024 on R&D program expense timing and as reductions to employee compensation end.

With all these factors in mind, our non-GAAP guidance for fiscal Q4 is as follows. We expect revenue to be \$3.9 billion plus or minus \$200 million; gross margin to be in the range of negative 10.5%, plus or minus 250 basis points; and operating expenses to be approximately \$845 million, plus or minus \$15 million.

We expect tax expenses of approximately \$40 million. Based on a share count of approximately 1.1 billion shares, we expect EPS to be a loss of \$1.19 plus or minus \$0.07.

In closing, we continue to act quickly and tenaciously to navigate this downturn, making the investments to maintain our leading capabilities across technology, products and manufacturing, while preserving our solid balance sheet. In this environment, we remain sharply focused on improving our profitability and free cash flow.

As market conditions improve, we will continue to drive efficiencies to hold on to productivity gains. Despite the impact of this downturn and effects of the CAC decision, we remain confident in our financial model and our ability to deliver long-term profitability, cash flow, and shareholder returns.

I will now turn it back over to Sanjay.

Sanjay Mehrotra {BIO 1977912 <GO>}

Thank you, Mark. I'm proud of the execution of the Micron team and the progress we've made this quarter. The leadership products we released and qualified are strengthening Micron's portfolio across multiple key markets. While there are near-term headwinds, I'm excited about the new product introductions that we have planned for the next several quarters, which will further enable us to leverage the dramatic growth in AI that is ahead of us.

I'm confident that this portfolio momentum, combined with our technology capability, manufacturing excellence, financial discipline, and excellent customer relationships, will position us well for the future. I also want to call attention to Micron's 2023 sustainability report, which published yesterday. The report underscores our continued commitment to innovation, the environment, our people, and the communities where we operate, outlining our progress and aspirations across our environmental, social, and governance programs. I encourage you to review the full report on Micron's website.

Thank you for joining us today. We will now open for questions.

Questions And Answers

Operator

(Question And Answer)

(Operator Instructions) And our first question comes from the line of C. J. Muse from Evercore ISI. Your question, please.

Q - C.J. Muse

Yes, good afternoon. Thank you for taking the question. I guess first question, with inventory expected to normalize in the coming months, quarters, how are you seeing customer purchasing behavior perhaps change, given that we're clearly hitting a pricing trough very soon? We'd love to hear kind of how those discussions might be changing.

A - Sanjay Mehrotra {[BIO 1977912 <GO>](#)}

Thanks, C. J., for that question. We of course, continue to work closely with our customers and as we said that customer inventories are improving. Except for data center, inventories are close to normal in most of our other end markets. Data center, we said by end of this year or somewhat thereafter -- shortly thereafter, data center customer inventories we expect to improve as well. And we continue to work closely with our customers. Some of the customers definitely interested in some of the longer-term outlook for the business and other customers operate on month-to-month basis.

And overall, of course, we continue to mitigate through some of the impact of the China decision as well. But the value that we are bringing to the customers for our products continues to strengthen and Micron is very much focused on navigating through the current downturn and working closely with our customers to address their future demand.

And as we have said in my remarks that some of the customers, given the low pricing that exists in the industry today and before prices begin to increase substantially, some of the customers may be looking at purchasing additional volumes at this time. But in general, the trajectory is of continuing improvement in their inventory levels, end-to-end across the supply chain, at the customers directly, as well as third parties who may be supplying to the customers, and as you know, inventories at the suppliers are coming down as well.

Q - C.J. Muse

Very helpful. If I could just follow up real quickly on HBM3, you guided to meaningful revs in fiscal '24. Can you give us a sense of what that means? And over time, what size kind of could that look like for you guys looking out kind of three, five years? Thank you.

A - Sanjay Mehrotra {[BIO 1977912 <GO>](#)}

Well, with respect to HBM3, we are very excited about this product. Micron has focused on bringing an industry-leading product, and HBM3 product that is in early stages of sampling and we expect to begin production volume ramp of this product in early 2024, is a product that has significantly higher performance, bandwidth, and significantly lower power. In fact, as a product, it is close to a generational leap ahead of anything else that is in the market.

We have received a strong endorsement for this product in the market, and we expect the volume ramp of this product for us to be rapid, to be steep ramp, and this will bring in, in our fiscal year 2024, strong revenue growth opportunity for us. So we are very excited about this standout product. It will be a significant growth driver for Micron.

And everything that we have done here is, of course, built on our industry-leading 1-beta technology, and applying to it, of course, advanced packaging, differentiated packaging, and TSV capabilities. So this is, we believe, going to be a standout product for us. And we expect -- we target a share with HBM with this kind of industry-leading product that would be higher than our average DRAM share in the industry as well.

Operator

Thank you. One moment for our next question. And our next question comes from the line of Timothy Arcuri from UBS. Your question, please.

Q - Timothy Arcuri {BIO 3824613 <GO>}

Thanks a lot. I had two. Mark, the first one is for you. I was wondering if you can sort of lay out what the fiscal Q4 guidance would have looked like, net of the ban. I know you said that that the impact from the ban gets worse actually in the fiscal first half. So is it as simple as maybe fiscal Q4, you had said low double-digit bit impact, but it sounds like it's probably not that big in fiscal Q4, so something less than that in fiscal Q4? And then you sort of expand to a number something in that range in the first half of fiscal 2025 -- '4, rather. Can you sort of handicap that for us and shape it for us?

A - Mark Murphy {BIO 16136048 <GO>}

Yes. We had a small impact in a -- very small in Q3. It's a more material impact in Q4. We do expect the impact to increase. However, our actions to mitigate that will help offset the effect. But really at this time, it's a headwind, but yes, there is -- and that's clear. However, we are taking mitigating actions and it's very uncertain, continues to evolve on what the impact will be. And again, the impact that we see in the fourth quarter is contemplated in our guidance.

Q - Timothy Arcuri {BIO 3824613 <GO>}

Got it. Maybe I'll ask you in the follow-up. But my second question is for Sanjay. So Sanjay, I asked you this last call too. So you alluded to the smartphone customers at least wanting to kind of get out in front of what they see maybe could be some tightness and maybe they're opportunistically trying to take advantage of pricing being so low. Can you just talk more broadly about what might change in your relationship with your customers coming out of this of this downturn?

I mean, could we be headed toward a situation where maybe the data center customers have pushed you and your peers so far during the downturn that maybe we can talk about LTAs at some point? I know that this is ways away, given kind of where we are today. But can you just talk about maybe over the past three months, when I asked you last time, how the tone of the discussion with the data center customers in particular has changed? Thanks.

A - Sanjay Mehrotra {BIO 1977912 <GO>}

Well, our customers, of course, work with us on LTAs. And as we have said, LTAs relate to their forecast for the year, generally. And while some customers may be operating on shorter term or other customers longer term, but generally speaking, they operate on yearly LTAs. And LTAs involve supply and demand commitments from the two sides. Of course, sometimes with the

changing industry environment on either side, on the supply or on the demand side, there can be adjustments made to those LTAs, and we work closely with our customers in those regards.

And we have had close relationships with the customers. We have a very strong product momentum. You are particularly inquiring about data center, and let me tell you that our product momentum in data center with strong portfolio of solutions, particularly addressing the growing interest in AI, in data center, generative AI, becoming a big opportunity, and we look at it for 2024 as a big year for AI and for memory and storage, and Micron will be well positioned with this product.

And these are all parts of our discussions when we address their requirements on their future purchases, when we address LTA requirements, and of course, we make the necessary investments related to our production mix in terms of die requirements, in terms of our assembly and test requirements, and we really work closely with our customers to help manage these.

And just keep in mind that, as I mentioned, that a lot of new product considerations go into the LTAs as well, as well as, of course, the volume and overall demand and supply considerations. So, LTAs, at the end, really help both the parties. They help us plan our engineering, our product roadmap, alignment on that, our investments in things such as back-end capacity, because products like HBM, products like high-density modules, and of course, in the mobile sector, products like MCPs, et cetera, have all different considerations at the back end and these are the kind of things LTAs really help us plan with our customers.

Q - Timothy Arcuri {[BIO 3824613](#) <GO>}

Thank you so much.

Operator

Thank you. One moment for our next question. And our next question comes from the line of Krish Sankar from TD Cowen. Your question please.

Q - Krish Sankar {[BIO 16151788](#) <GO>}

Yes. Thanks for taking my question and congrats on the good results. Sanjay, the first question I wanted to ask you was you spoke about AI servers having six to eight times more DRAM content and that demand is strong while traditional data center server demand is weak. There's a view that some of these AI servers are replacing over 10 of DCs -- regular DC servers. I'm just kind of curious how to think about overall DRAM demand as AI grows, but probably cannibalizes some of your regular data center server DRAM content. And then I had a follow-up.

A - Sanjay Mehrotra {[BIO 1977912](#) <GO>}

So, look, when we look at the overall DRAM demand, the DRAM TAM, of course, the AI is driving growth, automotive certainly driving growth. Other end markets, such as we mentioned mobile and PC, or consumer, in terms of their end demand has been somewhat lackluster.

The AI demand that is driven in data center, whether it is in the enterprise, definitely drives healthy trend for memory growth. Yes, enterprise server and some of the data center demand has been recently somewhat impacted by the macro trends, but the trend of AI and more memory is absolutely continuing. And that's what -- when we look at our overall 2023 demand growth and the projections of CAGR that we have ahead of us, we have taken those into account.

This is very, very early innings for AI and AI is really pervasive, it's everywhere in of course cloud applications, enterprise server applications, applications such as generative AI, would be in enterprises too because due to confidentiality of data, enterprises will be building their own large language models. And as you know, while the enterprise large language models may not be as large as the large language models you may see in examples such as superclusters, et cetera, but all of them are really tending toward greater number of parameters. Now we are talking about parameters with generative AI getting into even trillion parameter range; not too long ago, these used to be in 100 millions of range, that requires more memory.

So regardless of the applications, whether it is on the enterprise side or on the cloud server side, the memory requirements are continuing to increase. And I'll just point out that 6x to 8x that we have mentioned is the multiple of DRAM requirement in AI server versus standard server. And of course, as we highlighted in the script, there are many compute configurations, such as the supercluster example that we gave you, where the DRAM content that is required is few hundred times higher than a standard server.

So really, I think the journey here ahead of us will be very exciting. And when we look at machine-to-machine communication, when we look at opportunities for the virtuous cycle for ever-increasing data, that training applications, that inferencing at scale, and various edge applications, including automotive or driving, the requirements for memory and storage will continue to grow well, and Micron is going to be well positioned with our products. And we consider 2024 to be a big banner year for AI for memory and storage, and Micron will be well positioned to capture this with our strong portfolio of products from D5 to LP5 to HBM to high-density modules, even including graphics.

Q - Krish Sankar {[BIO 16151788](#) <GO>}

Got it. Very helpful, Sanjay. And then a follow-up for Mark. You said no inventory write-down in the current quarter expected, and if I remember right, Mark, you also mentioned in the past that inventory write-down is tied to your view on pricing three quarters out. So is it fair to assume that you're expecting a pricing drop pretty much this quarter?

And if the CAC decision does get really worse, that 15% to 25% of your sales get impacted, is there any more risk of inventory write-down or is that agnostic to the inventory write-down?
Thank you.

A - Mark Murphy {[BIO 16136048](#) <GO>}

Yes. Thanks, Krish. Maybe spend a few minutes just covering because it's a very complicated topic with a lot of moving parts, maybe spend a bit of time on the topic.

So our reported gross margin, our outlook, it's a function of many factors, including pricing. The inventory write-downs, which do include our -- or incorporate our forward view of pricing. The effects of utilization which you heard today, we've increased our -- or, reduced our wafer starts further. And then just volumes and associated leverage on period costs as discussed in previous quarters, and of course, mix.

These factors are continuously changing due to market environment and our actions. And as I've stated before, at these lower levels of profitability, our margin forecast and results are more sensitive to slight changes in assumptions such as price.

Now given price trends and our current view on pricing and costs, we took a material write-down in the second quarter, as we reported \$1.4 billion, took another \$400 million this quarter, and with

these write-downs, we've pulled forward inventory costs, thus lowered the carrying value of on-hand inventories.

As this lower-cost inventory clears in the future quarters, we realize more income in those quarters than we would have otherwise without the charge. So as an example, we took this \$400 million of additional write-downs in third quarter for inventories produced, and considering our latest views on volume mix, we also realized a benefit of near \$300 million from selling through the lower-cost inventories impacted by the second quarter write-down. So I do want to call out that it's with all the uncertainty, complexity, and sensitivity at these profitability levels, our write-down and the benefits that we had in the third quarter were not far off what we estimated in our guide.

So I think that's a good reflection of our handle of what's happening in the business. Now we've also got underutilization effects, which are creating higher costs and inventories and adding period costs. We project roughly \$1.1 billion of underutilization impact in FY '23 associated with the front end. Most of that will impact the P&L this year, some of it will carry over to next year, but because of the effect of the write-down accounting, less of it will carry over to next year than would have otherwise.

Beyond this period of write-down effects, the impact of lower wafer starts between the period costs and the higher cost inventories, the effect is higher single digits on margins than down to mid and lower single digits on margins as revenues increase.

So considering all this, just to give you a sense of profile of margin and in turn pricing to your question, we said last quarter that we expect -- or as we said last quarter, we expect -- we had a reported second quarter margin to be the trough, and that was driven by the \$1.4 billion write-down. With a much lower inventory charge forecasted in the third quarter which happened, that margin improved about 15 points.

And then also as mentioned last quarter, we said that fourth quarter would be better than third quarter on a lower write-down, and hence we guided today 5 points better than the third quarter. Again, these estimates are sensitive to pricing changes and -- but in our current view, we expect a gradual improvement on margin to continue sequentially on a reported basis.

Now if you take our non-GAAP third quarter gross margin of negative 16% and we were to strip out the write-down effects in third quarter, both the write-down portion and the realized benefit, and also to normalize, you strip out the insurance settlement which we had in the third quarter, we would still be -- those two things largely offset. So we'd be still at about 16% negative gross margin.

So again, over 100 net inventory effects, the 400 write-down, less than 300 realized benefit, and then roughly the same over 100 insurance settlement. So -- and this is a function of the pricing environment, which we -- I think properly captured in our guide. Now that adjusted margin 16% is down clearly versus the adjusted second quarter margin, which, as I recall, is about 7%, so down 23 points.

So under this adjusted view, we would trough on gross margin over the next few quarters, and then we would improve off these low levels through FY '24. So this is a profile that's consistent with what we've discussed before though the levels are a bit lower and a bit delayed. And so hopefully that provides you some color both on how we see pricing and how we see gross margin playing out with all the puts and takes.

Q - Krish Sankar {BIO 16151788 <GO>}

Yes. Thanks a lot, Mark. Thank you.

Operator

Thank you. One moment for our next question. And our next question comes from the line of Harlan Sur from J.P. Morgan. Your question, please.

Q - Harlan Sur {BIO 6539622 <GO>}

Hi, good afternoon. Thanks for taking my question. I guess as a follow-up to that, Mark, on your gross margin guidance for the fourth quarter, I know there are no inventory write-downs, but is it contemplating a step-up in underutilization charges or period costs associated with underutilization charges sequentially? And because you cut your wafer starts another 5 percentage points, right, to 30%, and if you could maybe quantify that step-up in underutilization charges.

And then as a follow-up, is the incremental 5% cut in utilization is primarily a result of the CAC restrictions?

A - Mark Murphy {BIO 16136048 <GO>}

It is not. It's more of an industry dynamic and our intent to get supply discipline in the market. Supply needs to come out of the market given inventory levels and that's the principal driver. As far as the effects of utilization, it is already incorporated in this guide. The period costs in the fourth quarter are about \$200 million and again, they're contemplated in the guidance.

Q - Harlan Sur {BIO 6539622 <GO>}

Perfect. Thank you.

Operator

Thank you. One moment for our next question. And our next question comes from the line of Ambrish Srivastava from BMO Capital Markets. Your question, please.

Q - Ambrish Srivastava {BIO 4109276 <GO>}

Hi. Thank you very much. Mark, I wanted to come back to the gross margin. When you had given the guidance for this quarter, you had walked us through in detail and you had said that stripping out all the adjustments and the inventory write-down, 3Q would be at 7.5%. Am I reading this right, that now you stripped out this negative 16%, right, so it's much worse than what you were thinking?

A - Mark Murphy {BIO 16136048 <GO>}

No. I don't think, it's much worse than what you're thinking. If you strip out just the underutilization effects, but you keep in the insurance settlement, you're close to what we said, sort of that 7% to 8%. So that you need to consider. We had said that was in there.

Q - Ambrish Srivastava {BIO 4109276 <GO>}

Got it. And then a follow-up either for you or for Sanjay, on the 15% round number, 15% of bit loss, share loss in China, how do you recoup that? Is that based on the assumption that bit growth or bit supply will be constrained? And so if the other two suppliers are able to meet the China

demand, they leave some demand out here in other regions for you to basically go after? Is there a pricing element to that? I'm just not very sure I understand how you get that.

A - Sanjay Mehrotra {[BIO 1977912](#) <GO>}

I will take that. So what we have said is that approximately 50% of our business in China is at risk of getting impacted. And of course, we are focused on mitigating any share loss with CIOs or as a result of CAC decision with those customers -- global customers who are not impacted by CAC decision.

So keep in mind that our share in DRAM is approximately 23% and our share in NAND is approximately 12%. So obviously, we have opportunities to gain share with other customers and this is what we are focused on. It will take some time and the CAC decisions can -- I mean, as we have said, it is hurting our business, it is slowing our recovery, it can result in quarter-to-quarter variations as well. But over longer term, our target is to maintain our share.

So while near term, CAC decision is challenging. Longer term, we will work with customers who are not impacted, our global customers, who are not impacted by CAC decision to increase our share. And we have a long history of working with our customers. We have brought tremendous value of our innovation, our supply, our product portfolio, supporting their innovation and roadmaps in the marketplace.

Our customers want to see a strong Micron. Our strategy of keeping our target share consistent over longer term with our current share is understood by our customers because again, they want to see a strong Micron. So they are supportive of this, and we will continue to work with our customers. And of course, as we bring value to our customers with our products and our product portfolio, we will focus on ROI on our investments and will certainly focus on improving the profitability of our business from current levels as well.

So we will of course keep profitability in mind, and again, it's important that Micron is a strong partner to our customers, and I think customers understand that multiple strong players in the industry is a benefit for multiple reasons to our customer ecosystem.

Q - Ambrish Srivastava {[BIO 4109276](#) <GO>}

Got it. Makes sense. Thank you, Sanjay.

Operator

Thank you. One moment for our next question. And our final question for today comes from the line of Tom O'Malley from Barclays. Your question, please.

Q - Tom O'Malley

Hey, guys. Thanks for taking my question. Recently, we've been picking up that there is a change in some of the A series, where you're starting to see some HBM2E used, just given the fact that there's limited capacity of HBM3. I guess part one is, are you seeing an ability to service that market today?

And then the second part of the question is, you're saying that AI servers see about 6x to 8x DRAM content. I assume that contemplates some HBM, but you guys were talking about some AI tailwinds today, when you're really not servicing that market as much. So could you talk about what you're seeing ex-HBM as the multiplier effect on DRAM today, just so we can get a picture

of how you guys are seeing the improvement in data center with AI today, ex that product? That'd be really helpful. Thank you.

A - Sanjay Mehrotra {[BIO](#) [1977912](#) <GO>}

So certainly, we have had HBM2E product in the marketplace that actually gave us strong experience in bringing up our technology and production capability with HBM. The market, as I mentioned, have shifted -- is shifting to HBM3, and Micron's HBM3 plus product, which I called as a generational leap ahead of anything in the industry, is going to position us well as we bring that into volume production during the course of our fiscal year '24, starting early part of calendar '24, contributing to several hundred million dollars of revenue opportunity over time.

And with respect to AI part of the market, I want to be very clear that yes, with respect to high-density modules and with respect to high-bandwidth HBM3 solutions, that part of the market is growing this year, and it's an opportunity that we want to capture, and I believe that we'll be well positioned to capture, as I mentioned, that we will be targeting share in HBM with our absolute industry-leading product that's higher than our DRAM industry average share.

So -- but it's important to understand is that AI is being served not only by HBM or high-density DRAM modules, but it is also being served by D5 memory and by LPDRAM as well. And this is where with D5 and LPDRAM products, we gave you some examples in our script as well, large amount of LPDRAM being used in industry-leading, high-performance compute platforms.

In fact, the 144 terabytes that we mentioned in DGX GH200, about 122 terabyte of that is LPDRAM. And Micron is very well positioned with differentiated solution of our LPDRAM there today. So I think it's important to understand that the AI server market is made up of HBM, it's made up of high-density DRAM modules, includes -- it also is made up of DDR5, LP5 and some element of graphics memory as well.

So we do have a broad portfolio, and in 2024 with HBM and high-density DRAM modules getting into production, I really believe we'll be extremely well positioned to capture the growing opportunity in AI. And 75% of DRAM on AI servers today is DDR5. And as I emphasized and as I'm sure you well know, we participate very well in D5. In fact, we led the industry with our D5 products, again, built on 1-beta technology here.

Q - Tom O'Malley

Thank you, Sanjay, and I appreciate you guys sneaking me in.

Operator

Thank you. This does conclude the question-and-answer session as well as today's program. Thank you, ladies and gentlemen, for your participation. You may now disconnect. Good day.

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